HWC-SR501 PIR Sensor Module

Specification



Based on infrared technology automatic control products, HC-SR501 has high sensitivity and reliability, and is widely used in various types of automatic induction appliances, such as automatic doors, automatic washing mobile phones, automatic liquid droppers, sensor flushers, various types of smart toys, energy-saving lamps. , Automatic body-sensing night light and other induction electrical equipment.

Features

1 Fully automatic sensing: When a person enters its sensing range, it outputs a high level. When the person leaves the sensing range, it automatically delays and turns off the high level and outputs a low level.
2 Photosensitive control, no induction in daytime or strong light, night induction. (Default is no photosensitive control)
3 two trigger modes (default repeatable trigger)
a. Non-repeatable triggering mode: After the sensor output is high, once the delay period ends, the output will automatically change from high level to low level;
b. Repeatable triggering mode: after the sensor output is high, during the delay time period, if there is activity of the human body in its sensing range, the output will remain high until the person will delay and then leave high power. Level changes to low level (the sensing module automatically delays a delay period after detecting each activity of the human body, and the time of the last activity is the starting point of the delay time).

Technical Parameters

Product Model	HC-SR501	Batch can be customized	● The level of the output is not an absolute value. The change of the ambient temperature will have a slight influence on it, and an error value of about ±0.5V may occur. ● Modules are reserved CDS photoresistor interface for light control functions; ● Other customized items may add additional costs. It is recommended to maintain regular parameters. Please discuss and communicate with relevant staff for specific matters.
Working Voltage	DC4.8V-20V	can be customized	
Static Power	<50 uA;		
Level Output	Induction3.3V No induction0V	can be customized	
Delay Time	0.5-200S	Available fixed resistance	
Block Time	2.5s	can be customized	
Trigger Mode	L can not be repeated, H can be repeated default value H		
Induction Distance	5、8、10m	can be customized	
Induction Angle Range	100°	can be customized	
Working Temperature	-20—75℃		
Size	32 x 24 x 7mm	can be customized	

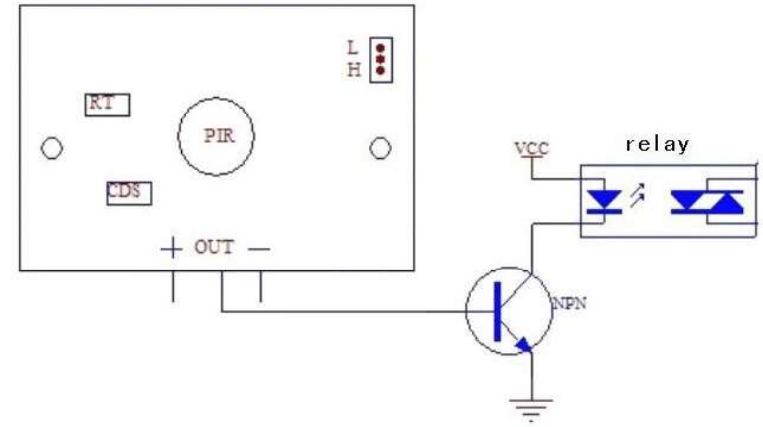
Precautions:

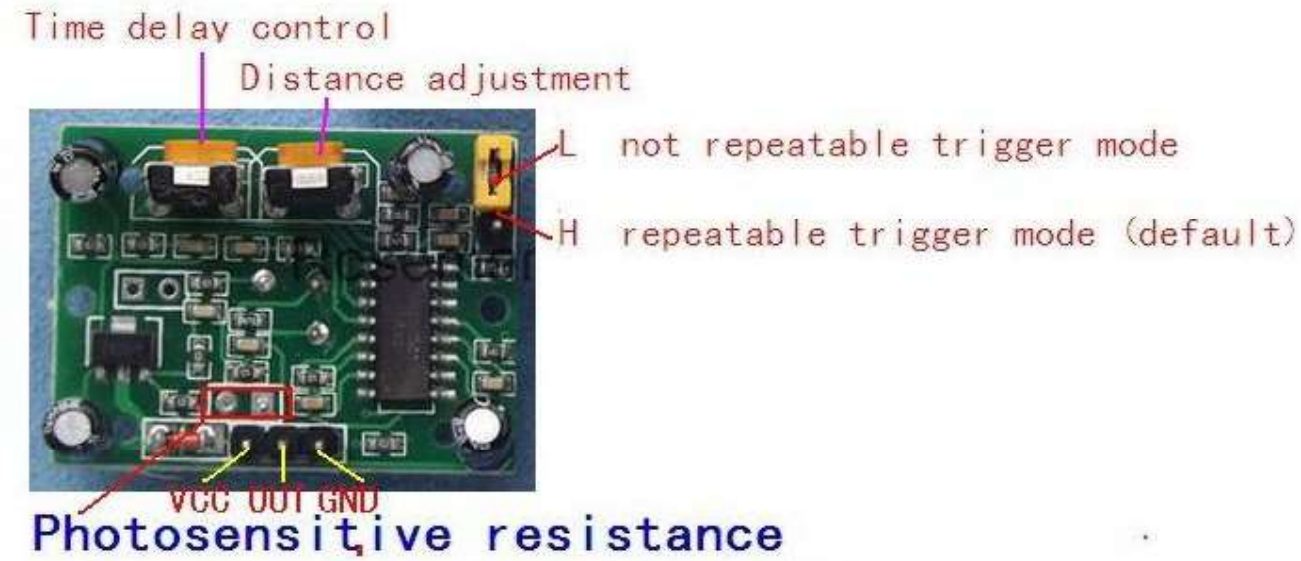
● The pyroelectric infrared sensor should be 2.0-2.2 meters away from the ground. Keep away from air conditioners, refrigerators, stoves and other sensitive places where air temperature changes.
● There must be no screens, furniture, large bonsai or other partitions within the detection range.
● The pyroelectric infrared sensor should not be directly facing the window, otherwise the interference of hot air

currents outside the window and the movement of people will cause false alarms. It is better to have the curtains on conditional conditions. Infrared pyroelectric sensors should also not be installed where there is strong airflow.

● The sensitivity of the pyroelectric infrared sensor to the human body is also related to the direction of human motion. Pyroelectric infrared sensors are the least sensitive to radial shift reactions, and are most sensitive to moving sides in the cross-cut direction (ie, the direction perpendicular to the radius). Selecting the proper installation location at the site is to avoid false alarms from the infrared probe and to obtain the best detection sensitivity.

External Schematic





Photosensitive resistance

1. Adjust distance potentiometer clockwise rotation, response distance increases (about 7 m); On the other hand, response distance decrease (about 3 m).

2. Adjust delay potentiometer clockwise rotation, response time delay increase (about 300 s); On the other hand, response time delay reduced (about 0.5 s)

Physical picture



Commonly used lens selection table:



Model: 8002-2B
Angle: 100°
Distance: 10m
Size:φ23mm



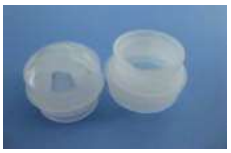
Model: 8003-3
Angle: 120°
Distance: 10m
Size: φ25mm



Model:8090
Angle: 120°
Distance: 8m
Size:φ17.8mm



Model: 8120
Angle: 120°
Distance: 5m
Size: φ12.7mm



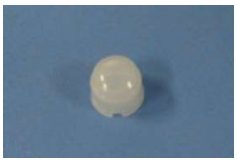
Model: 8120-5
Angle: 120°
Distance: 6m
Size: φ12.7mm



Model: 8308-3
Angle: 120°
Distance: 5m
Size:φ11.8mm



Model: 8308-5
Angle: 90°
Distance: 5m
Size: φ10.27mm



Model: 8308-5B
Angle: 90°
Distance: 5m
Size: φ10.27mm

Instructions for use:

1. The induction module has an initialization time of about one minute after power-on. During this period, the module will output 0-3 times at intervals, and enter the standby state after one minute.

2. Try to avoid the interference of lights and other interference sources on the lens of the module surface at close range, so as to avoid the introduction of interference signals to cause misoperation; the use environment should try to avoid flowing wind, which will also cause interference to the sensor.

3. The sensor module adopts dual element probe, the window of the probe is rectangular, and the dual element (A element and B element) is located at both ends in the longer direction. When the human body walks from left to right or from right to left, the infrared spectrum reaches the dual element There is a difference in time and distance. The larger the difference, the more sensitive the induction. When the human body walks from the front to the probe or from top to bottom or from bottom to top, the dual element cannot detect the change of the infrared spectrum distance, there is no difference, so the induction is not sensitive or does not work.

4. Adjust the distance to turn the potentiometer clockwise, the sensing distance increases (about 7 meters), otherwise, the sensing distance decreases (about 3 meters).

Adjust the delay potentiometer to rotate clockwise, the induction delay will increase (about 300S), otherwise, the induction delay will decrease (about 5S).